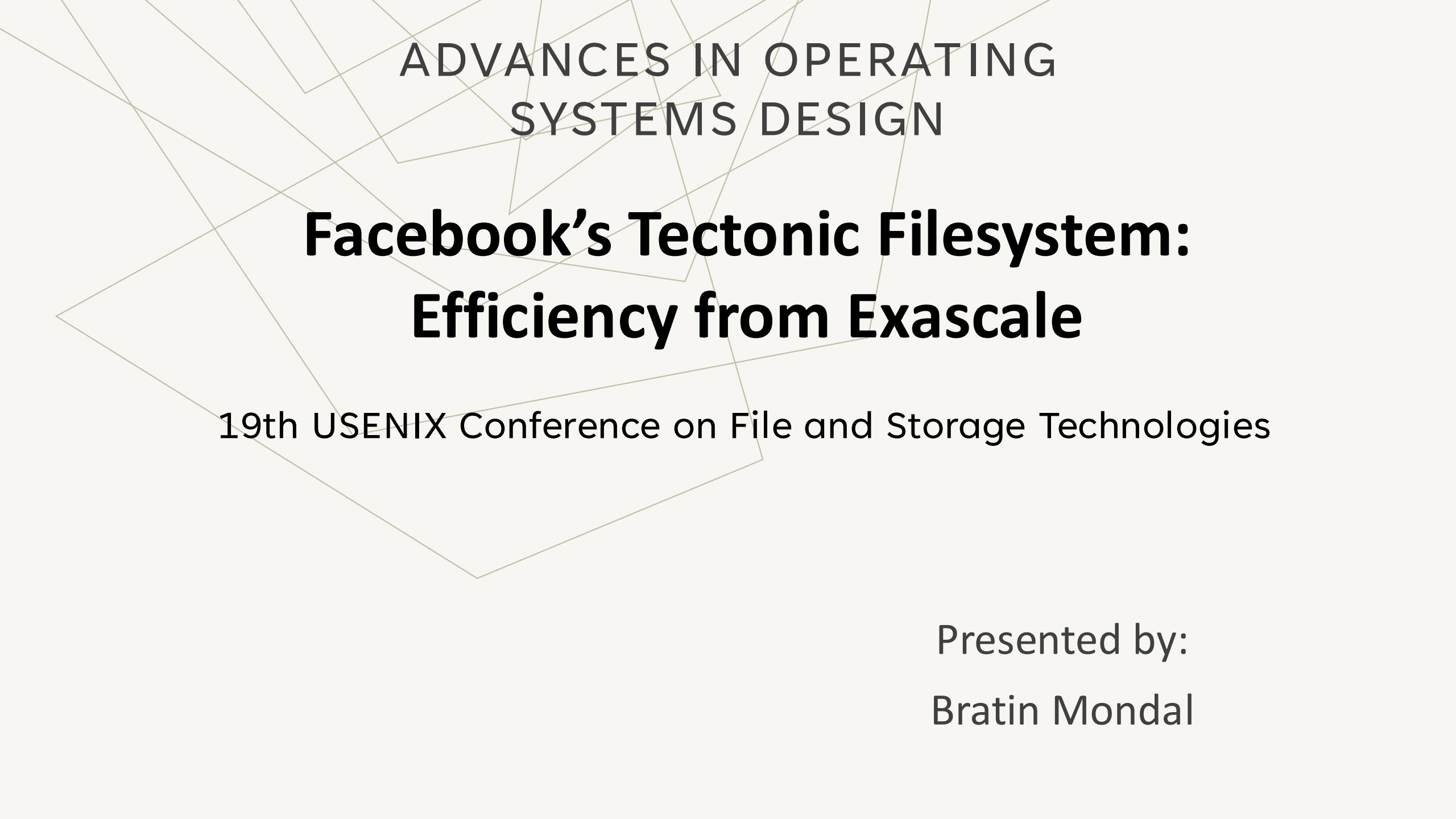




ADVANCES IN OPERATING
SYSTEMS DESIGN

Facebook's Tectonic Filesystem: Efficiency from Exascale

19th USENIX Conference on File and Storage Technologies

Presented by:
Bratin Mondal

Overview: What is Tectonic ?

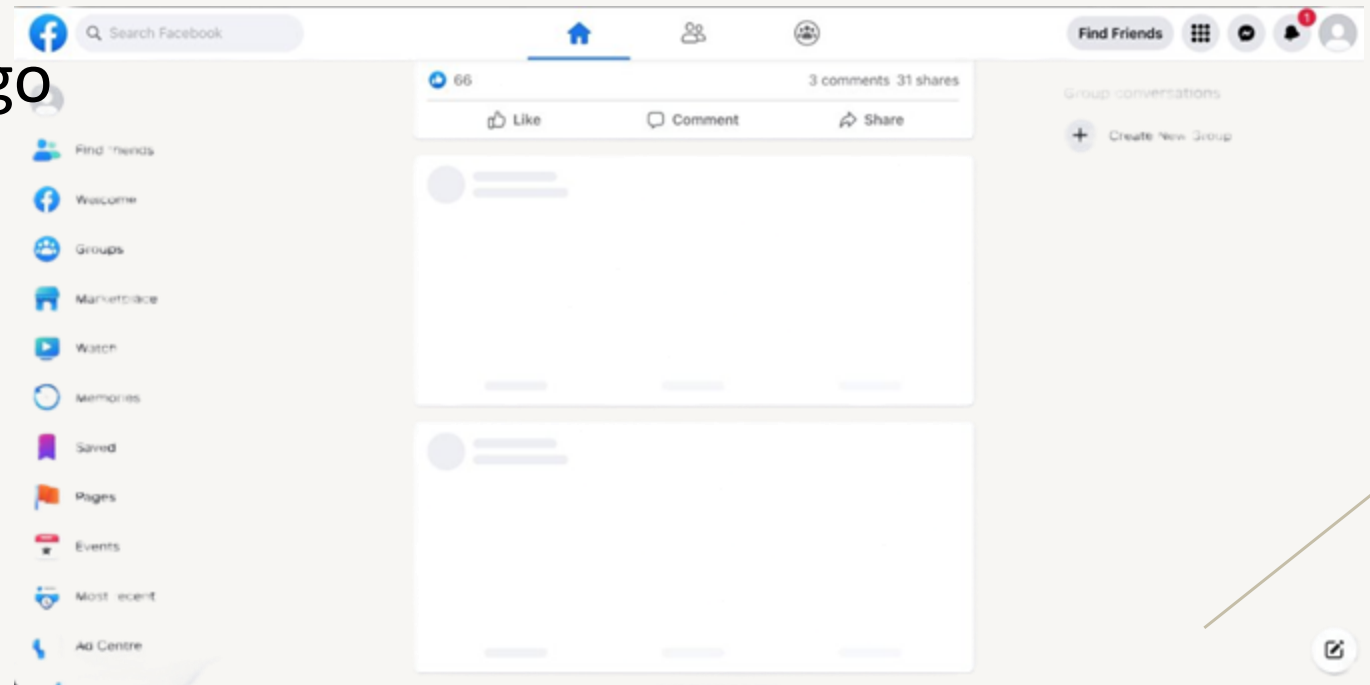
- Meta(Facebook)'s **exabyte-scale** distributed filesystem
- Serves around ten tenants with different requirements
 - Tenants are distributed systems that do not share data with each other and having **different set of requirements**
- Not optimized for any particular workload but designed in such a way that **tenants can enable their optimizations**
- Having various types of tenants enable sharing of surplus resources
 - Some tenants need IOPS, some need disk throughput

Meta's Storage Infrastructure before Tectonic

- Tenants used specialized storage systems
- Majorly two types of tenants
 - Blob Storage
 - Data Warehouse

Blob (Binary Large Objects) Storage

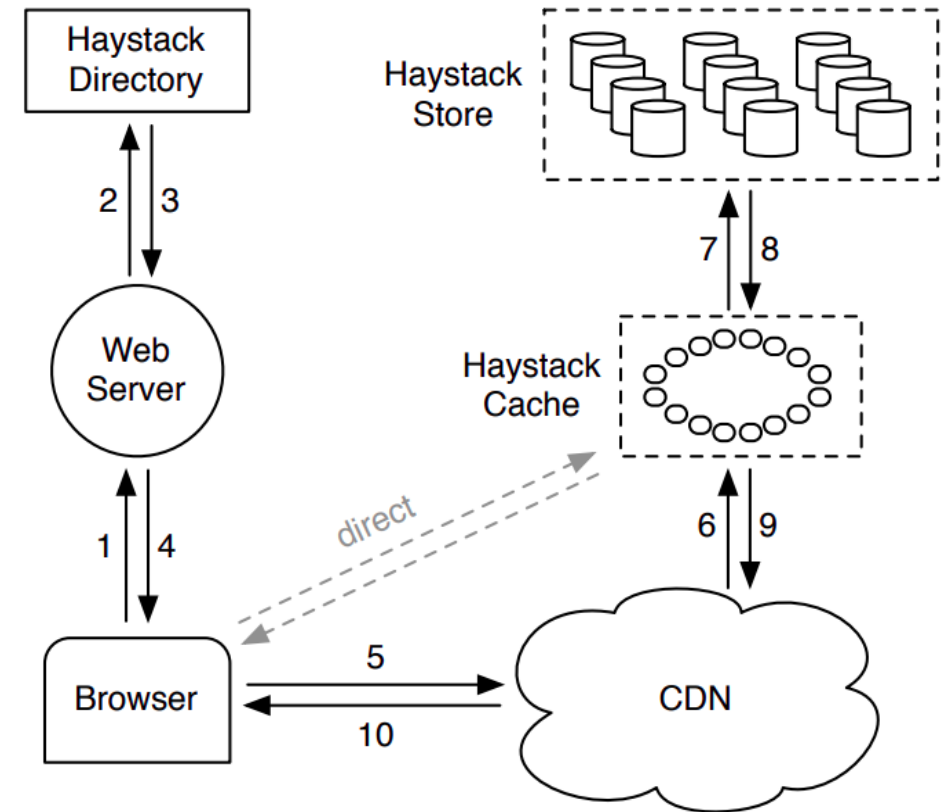
- Immutable, varies from kilobytes(small photos) to megabytes(HD Videos)
- Low-latency reads and writes required
- Hot Blobs
 - Photo uploaded one minute ago
- Warm Blobs
 - Photo uploaded one year ago



Haystack – Hot blob storage

`http://<CDN>/<Cache>/<Machine id>/<Logical volume, Photo>`

- Minimize the number of disk accesses
- In-memory metadata
- Collapse many images together into a single file
- Replication factor – 3.6x (**Ideal**)
 - Each logical byte is replicated 3 times physically
 - Additional 1.2x overhead for RAID-6 Storage
- **Issue?**
 - Haystack needed very high IOPS
 - Need more devices to meet the IOPS requirements
 - Spare disk storage capacity
 - Replication factor **5.3x** in practice
 - Can we use the surplus storage for others?



f4 – Warm blob storage

- Storage efficiency without compromising fault tolerance
- Uses RS(10,4) encoding
- Each logical byte is stored in two different datacenters
- Replication factor of 2.8x

- Blobs shifted from hot storage to warm storage

- **Issue?**

- Large number of storage nodes needed to store the vast amount of warm blobs
- But it also provides IOPS capacity which is not needed for the less frequently-accessed data
 - Can we use the surplus IOPS for others?

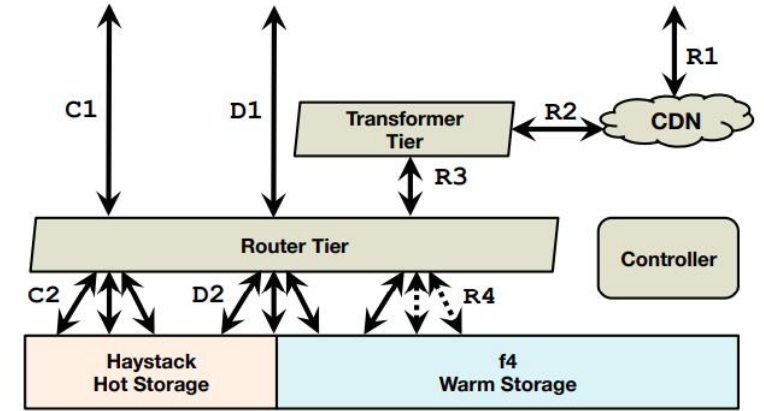
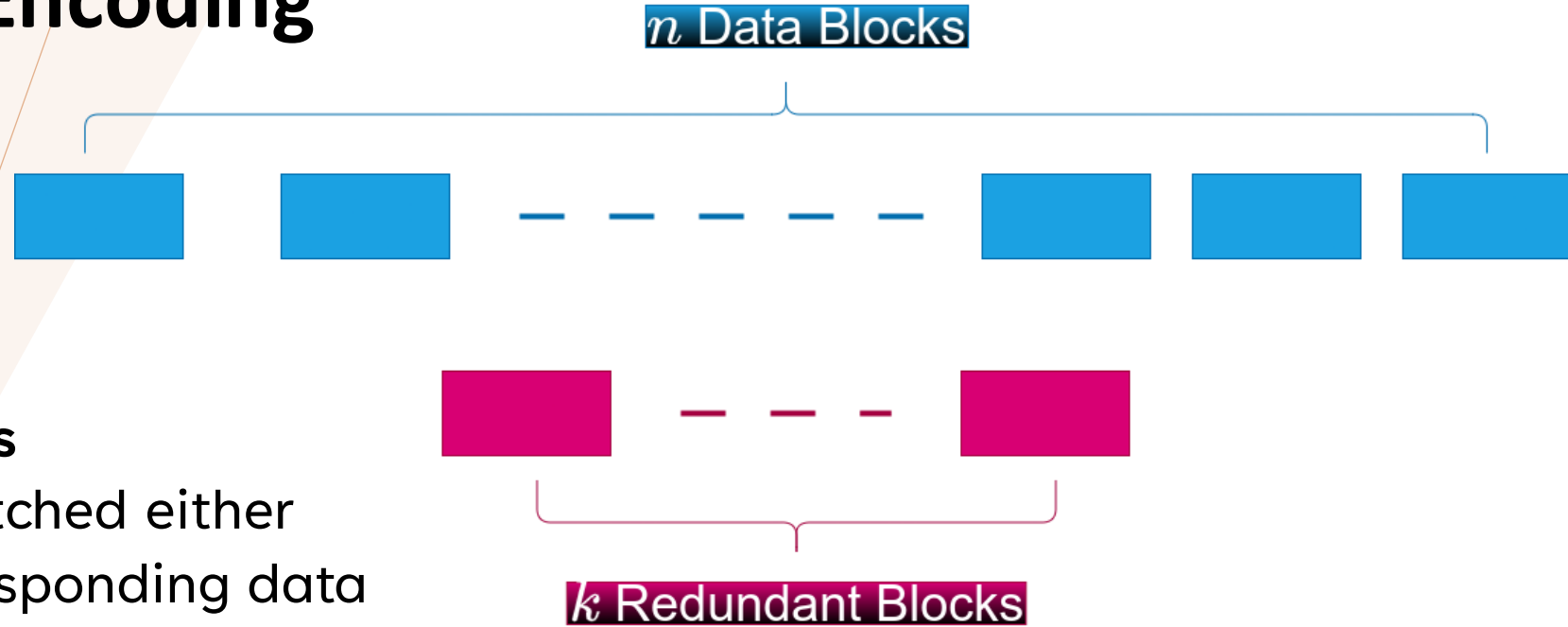


Figure 6: Overall BLOB Storage Architecture with creates (C1–C2), deletes (D1–D2), and reads (R1–R4). Creates are handled by Haystack, most deletes are handled by Haystack, reads are handled by either Haystack or f4.

Reed-Solomon(RS) Encoding

- Based on polynomial error-correcting codes over finite fields
- RS(n,k) can tolerate **maximum of k block faults**
- Each data block can be fetched either
 - Directly from the corresponding data block
 - Reconstructed using the redundant block and other data blocks

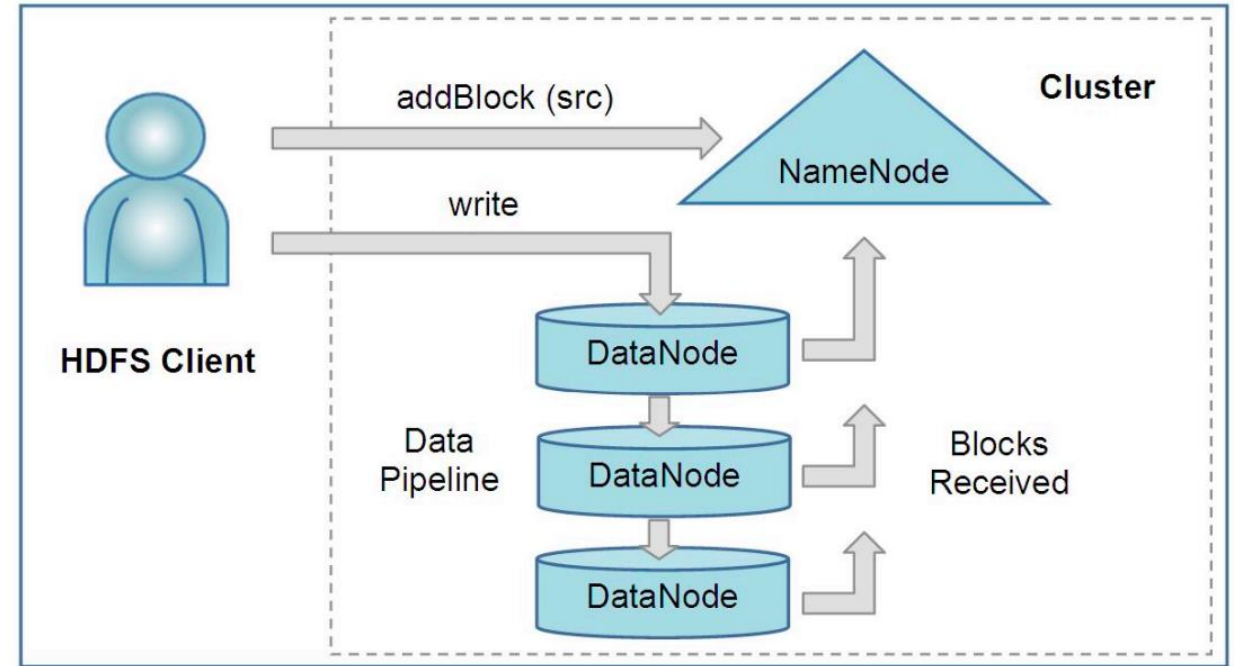


Data Warehouse

- Storage for data analytics (map-reduce tables, snapshots of social graph, etc.)
- Read/Write throughput prioritized over latency (Batch processing)
- Reads averaging multiple megabytes
- Writes averaging tens of megabytes

Hadoop Distributed File System(HDFS)

- NameNode to store and serve metadata
- DataNode to serve read/writes
- **Issue?**
 - HDFS clusters are limited in size because of single NameNode
 - Data needs to be distributed among multiple clusters
 - 2D Bin Packing (NP-Hard) -
 - Cluster capacity
 - Cluster throughput



Challenges

1. Scale to exabyte scale

- How to store the large amount of metadata?

2. Performance isolation between tenants

- In a storage system used by a large number of tenants, how to prevent one tenant's performance from being affected by others?

3. Tenant-specific optimizations

- In a generalized filesystem, how to provide tenants with support that meets the guarantees typically offered by specialized filesystems?

Some existing Systems

- Federated HDFS and Windows Azure Storage (WAS)
 - Merge multiple smaller storage clusters into larger clusters
 - Multiple independent namespaces but Data nodes are shared – Still bin-packing at namespace level (In which namespace to put what data?)
- Ceph and Fault-tolerant Distributed Storage(FDS)
 - Hash Data object to get data location – Increase the Range of hash function to scale
 - Hash function needs to be updated on each data relocation(Too costly in large systems)
- For multitenancy, there exists a few complex resource management techniques to provide optimal fair sharing
- (We will see some filesystems with similar design as tectonic at the end)

Tectonic Cluster

- Durable storage, resilient to host, rack and power domain failure
 - Tenants can build geo-replication for datacenter failures
 - Supports any number of arbitrarily-size namespaces or filesystem directory hierarchies
-
- Storage Nodes
 - Metadata Nodes
 - Stateless Nodes for background operations
 - Client library interacts with tectonic

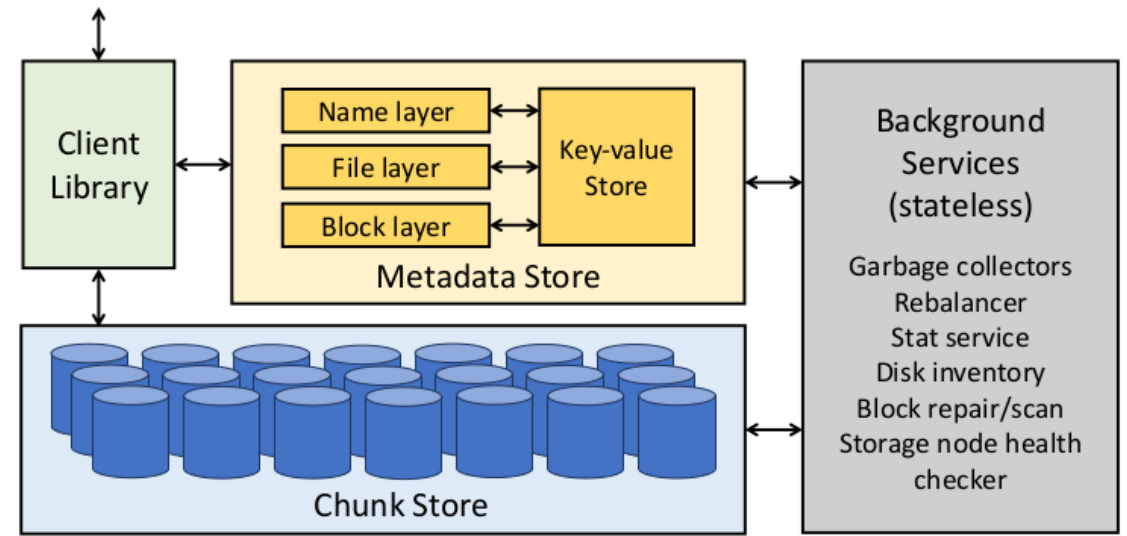


Figure 2: Tectonic architecture. Arrows indicate network calls. Tectonic stores filesystem metadata in a key-value store. Apart from the Chunk and Metadata Stores, all components are stateless.

Chunk Store

- Files divided into blocks and blocks are divided into chunks
- Abstraction created by the client library using metadata
- Chunks are stored as files in cluster's storage nodes using XFS
 1. Scalable – Linearly grows with the number of storage nodes
 2. Read/write to storage nodes can be specialized to tenant's performance needs

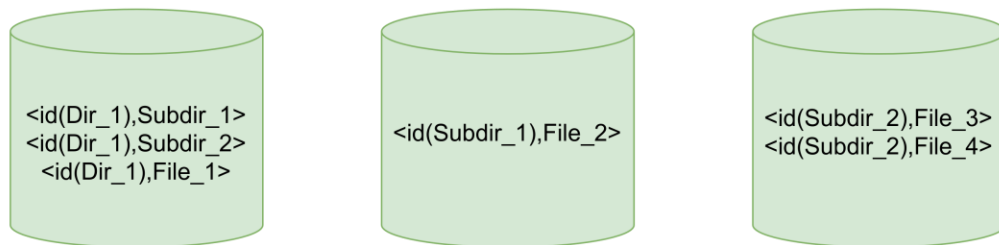
Metadata Store

Layer	Key	Value	Sharded by	Mapping
Name	(dir_id, <i>subdirname</i>)	subdir_info, subdir_id	dir_id	dir → list of subdirs (expanded)
	(dir_id, <i>filename</i>)	file_info, file_id	dir_id	dir → list of files (expanded)
File	(file_id, blk_id)	blk_info	file_id	file → list of blocks (expanded)
Block	blk_id	list<disk_id>	blk_id	block → list of disks (i.e., chunks)
	(disk_id, blk_id)	chunk_info	blk_id	disk → list of blocks (expanded)

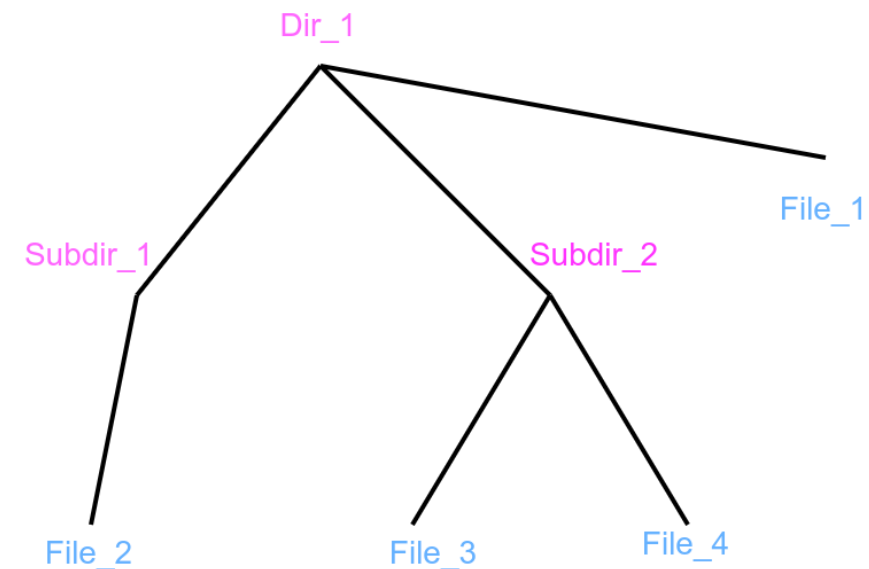
Table 1: Tectonic’s layered metadata schema. *dirname* and *filename* are application-exposed strings. *dir_id*, *file_id*, and *block_id* are internal object references. Most mappings are expanded for efficient updating.

- Three metadata layers
 - Name Layer: Directory → Sub-directories/files
 - File Layer: File → Blocks
 - Block Layer: Block → List of disks(chunks), Disk → Block
- Sharded by Key using Hash Partitioning
- Stored in expanded format (Fast updates)

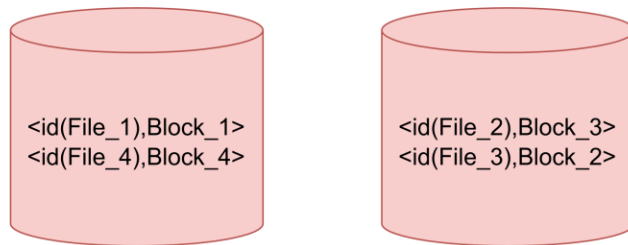
Metadata Storage



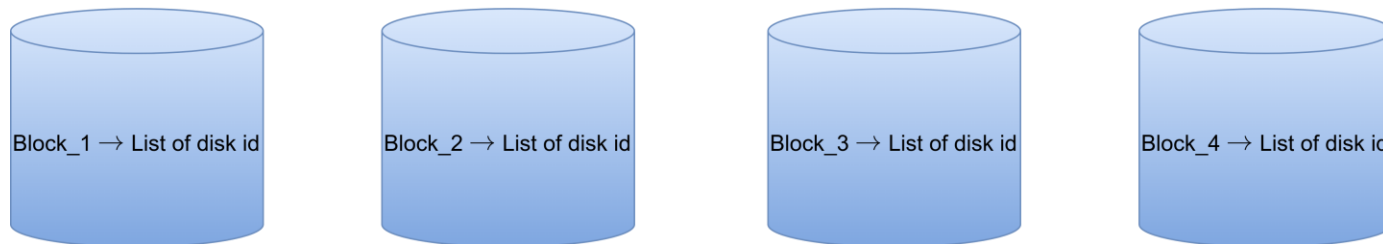
Name Layer



Namespace



File Layer



Block Layer

- Managed by ZippyDB at a shard granularity
- Metadata store nodes internally runs RocksDB and shards are replicated using Paxos
- Advantages
 1. Scalable – Metadata storage grows with number of metadata nodes
 2. Layered metadata approach and hash-partitioning avoids hotspots
- Issue
 1. Higher latency
 - Cache Name and File layer once sealed
 2. No support for atomic cross-directory move operations

Client Library

- Provides filesystem abstraction to applications
- Client library operates read/write operations at chunk granularity

Single Writer Semantics

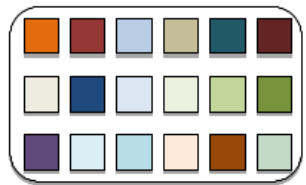
- Token provided to client when it opens the file for appending
- Updates to metadata must contain the token
- Last process having opened the file is the only writer
- Tenants needing multiple-writer semantics can build their own serialization semantics on top of tectonic

Background Services

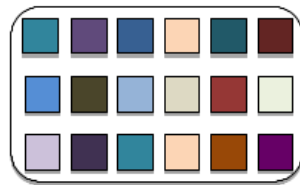
- Operate on one shard at a time
- Garbage collector
 - Removes inconsistencies between metadata layers
 - From multi-step client library operations
 - From lazy object deletion
- Rebalance and Repair
 - Identify lost chunks and repair them
 - Uses the reverse index mapping from disks to blocks to scale horizontally

Copysets

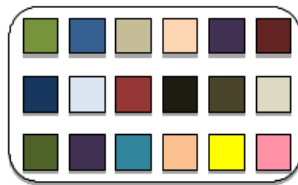
- Set of nodes for which simultaneous crash will lead to data loss
- What if we put copies of data in randomly chosen storage nodes?



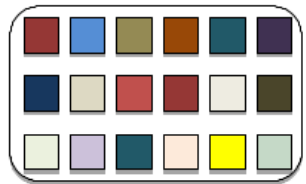
Node 1



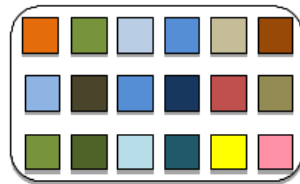
Node 2



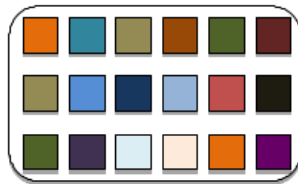
Node 3



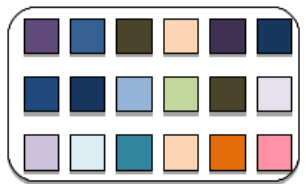
Node 4



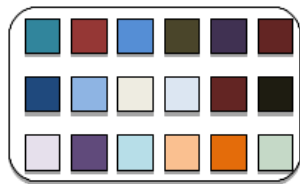
Node 5



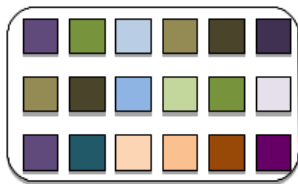
Node 6



Node 7



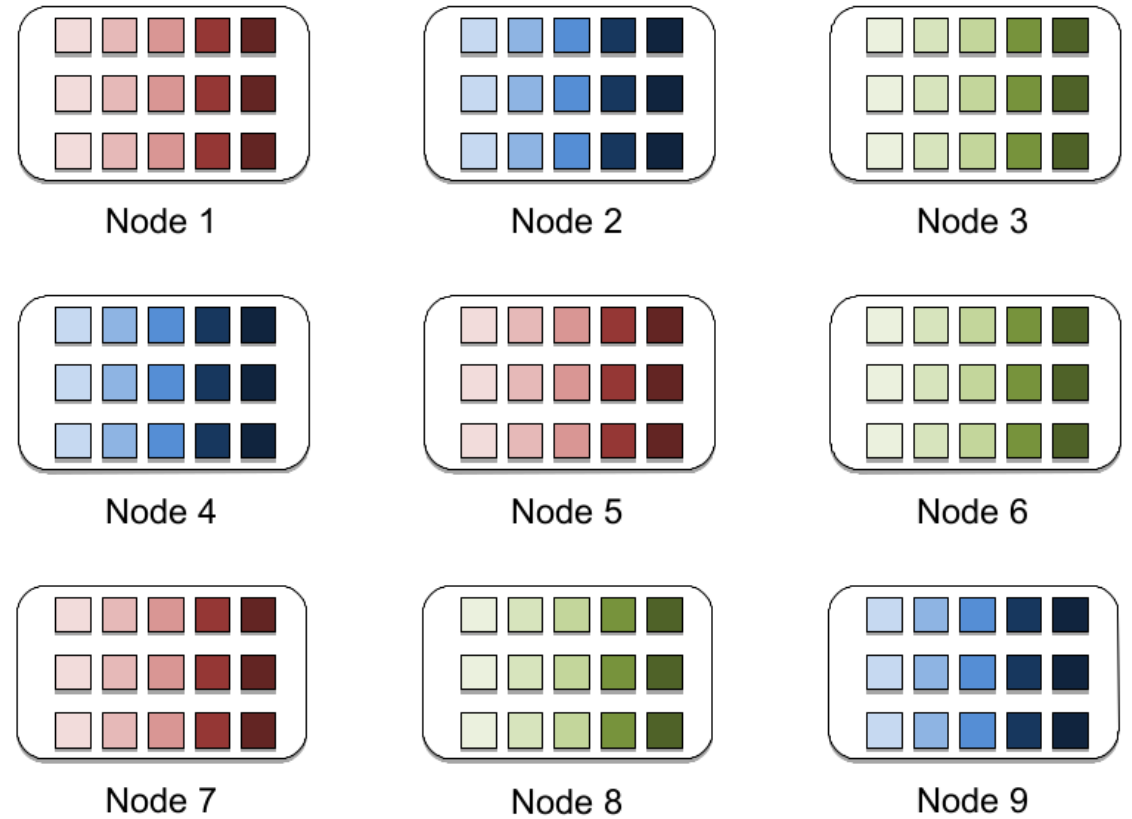
Node 8



Node 9

- Huge number of copysets
- Data is lost if any three of the nodes go down

- Only 3 copysets - $\{1,5,7\}$, $\{2,4,9\}$, $\{3,6,8\}$
- Probability of data loss is very low
- **What is the issue?**
- We lose data very less frequently, but we would lose large amount of data
- Reconstruction can be done only from the fellow nodes in the copyset



- Scatter Width – Number of nodes from which reconstruction can be done
- Create random permutations and group consecutive disks in a copyset
- While writing a block, choose a primary node randomly
- Choose a permutation corresponding to a block id, and use the copyset for the primary node



Permutation 1: Scatter width = 2



Permutation 2: Scatter width = 4



- Reconstruction load is now distributed
- In tectonic, the block layer and the rebalancer services together attempt to maintain a fixed copyset count.

Multitenancy

- Non-ephemeral resource: Predictable and slow changes
 - Storage Capacity: Pre-allocated capacity (No dynamic adaptation)
- Ephemeral resource: Real-time automated management
 - IOPS and Metadata Query
- At what granularity to manage?
 - Tenants?
 - Too much generalization – Tenants have different applications with different workloads
 - Applications?
 - Too complex and resource intensive

- **TrafficGroups and TrafficClass**
 - Applications within same Trafficgroup have similar latency and resource requirements
 - TrafficClass – **Gold**(Latency-Sensitive), **Silver**(Normal), **Bronze**(Background applications)
- **Global Resource Sharing**
 - **Aim:**
 - Limit resource usage
 - Share surplus resources effectively
 - Modified Leaky bucket algorithm – High-Performance, Near-Realtime Distributed Counters
 - Increment the counter – If spare capacity, use it
 - Check in other TrafficClass in same tenant
 - Check in other tenants
 - Using other TrafficClass resource – Resulting Traffic Class gets the minimum class

- **Local Resource Sharing at Metadata and Storage Nodes**
 - **Aim:**
 - Avoid it being hotspot
 - Meet latency requirement for Gold class
 - Weighted Round-Robin(WRR): Skip already if quota is used or will exceed if granted
 - Some optimizations:
 - Lower request class can give its turn to higher request if it will have enough time after the higher request gets serviced (Similar to SJF)
 - Limit number of non-Gold traffic in the queue for a disk if there are pending Gold requests
 - **Disks may re-arrange requests!** Stop scheduling non-gold requests if some gold request is pending for some threshold amount of time
- **Note:** Resource sharing need not be done again at Nodes because you have already done it globally

Access Control

- Token Based Access Control Mechanism
- Each Layer give the token for the next layer
- Tokens are piggybacked

Tenant Specific Optimizations

- **Data Warehouse Write Optimizations** – Read only after complete write semantics
 1. **Asynchronous writes:**
 - Applications themselves buffer writes until block size data available
 - RS-encode (less memory and network overhead)
 - Write the chunks in parallel
 - No inconsistency since metadata only update after complete write
 2. **Hedged Quorum Writes:**
 - First send reservation requests to more than the numbers of nodes needed
 - Don't wait for successful completion of writes to all the nodes. Wait only until **majority** of the blocks have been written and then reply to client.
 - Improves 99th percentile latency

- **Blob Storage Optimizations –**
 - 1. Consistent partial block appends:**
 - Wait only till quorum size appends
 - Only block creator can create append
 - Metadata updated with post-append block size and checksum
 - 2. Reencoding blocks for storage efficiency:**
 - Enabled by Client Library-driven design. Generally filesystems need you to preconfigure the way you want to store files.
 - From replicated to RS encoding

Tectonic in Production: Observations and Lessons

- Performance comparable to specialized filesystems
- Data warehouse request spikes handled by surplus resources
- Only name layer metadata was seen to be hotspot (1% time)
 - Master uses the metadata using list API and informs the worker nodes
- Reconstruction Storm:
 - Restrict reconstruction load to 10%(**tuned**): Contiguous vs Striped
- Client Library access data directly:
 - Alternative design would use a frontend proxy – Avoid an extra network node
 - But bugs in library become bug in application

Tradeoff

- Higher Metadata Latency
 - As such no direct solution
 - Applications need to adjust how certain metadata operations are handled
 - Parallelize if possible
- Hash partitioned metadata – avoids hotspot
 - No recursive list API – Client needs to build a wrapper for it
 - You don't have updated directory usage information (updated periodically)
 - Commands like du(directory utilization) not available

Conclusion

- Not every service at Meta use Tectonic (Graph storage, Key-Value Storage)
- Many services communicate with tenants which use Tectonic underlying

A series of thin, light brown lines forming an abstract geometric pattern in the top left corner of the slide. The lines intersect to create various triangular and polygonal shapes.

THANK YOU

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